

MelodyMap: Urban Music Space and Night-Time Activity

Dingming Zheng

Abstract

Live-music venues are part of the cultural and commercial life of urban neighbourhoods, but their relationship with surrounding night-time activity is difficult to interpret from spatial data alone. This project studies that relationship using Melbourne open data and controlled simulation analysis. A real central-Melbourne case study first shows that a music venue can be close to several pedestrian sensors, so the measured local activity depends on how spatial matching is defined. The broader analysis then combines venue concentration, hourly pedestrian activity, music-district structure, and nearby hospitality capacity. Simulation analysis tests how the observed relationship changes with sensor coverage, spatial radius, coordinate error, urban centrality, and the strength of a direct music effect. The results show a clear spatial association between music concentration and night-time activity, while also demonstrating that shared centrality can generate a similarly strong relationship even when the direct music effect is set to zero.

1 Background

Live music is closely connected to urban night-time life. Music venues often appear near restaurants, bars, transport hubs, and busy pedestrian streets, creating neighbourhoods where cultural activity and consumption opportunities overlap. Mapping these places can help describe where music districts form and how they relate to the wider activity of the city. The main analytical challenge lies in interpreting the observed relationship. Busy central areas may contain both more music venues and more pedestrians simply because they are well connected, commercially dense, and highly accessible. Spatial measurement choices can also affect the results: several sensors may plausibly represent the same venue, while changing the analysis radius alters which venues are considered nearby. To address these issues, the project combines a real Melbourne case study with controlled simulation analysis. The real data provide the empirical urban context, while the simulations examine when the observed relationship remains stable and when it may instead reflect measurement choices or underlying spatial structure.

2 Data and Method

The analysis uses City of Melbourne information on live-music venues and pedestrian counting locations, together with hospitality indicators describing restaurant seating and bar or pub capacity. Pedestrian movement is used as an indicator of urban activity rather than direct spending, while hospitality capacity describes the surrounding consumption environment.

For the real case study, I calculate Haversine distances between selected music venues and pedestrian sensors. I use 500 m as the main local-neighbourhood scale and vary this radius later in the simulation analysis. I summarise night-time activity over the 18:00–23:00 period, identify music districts through spatial clustering, and compare local music density with pedestrian activity. The simulation analysis models pedestrian activity at sensor i as

$$F_i = \beta M_i + \gamma C_i + \varepsilon_i, \quad (1)$$

where M_i denotes local music density, C_i urban centrality, β the direct music effect, γ the influence of shared centrality, and ε_i random variation. I vary sensor coverage, spatial radius, coordinate noise, centrality confounding, and direct effect size. For each parameter setting, I run 30 simulations, producing 630 runs across 21 conditions. I then use Spearman correlation to quantify the recovered association between local music density and pedestrian activity.

3 Real Melbourne Case Study

Table 1: Distance from each selected live-music venue to each pedestrian sensor (metres).

Venue	Bourke Street Mall (S)	Flinders St. Underpass	Town Hall (W)
Curtin House	203	795	333
Sticky Institute	451	153	325
The Sub Club	405	221	319

Table 1 shows that the selected venues do not map cleanly to one pedestrian sensor. At a 500 m radius, eight of the nine venue-sensor pairs remain plausible. This makes the matching rule part of the analysis rather than a minor technical choice.

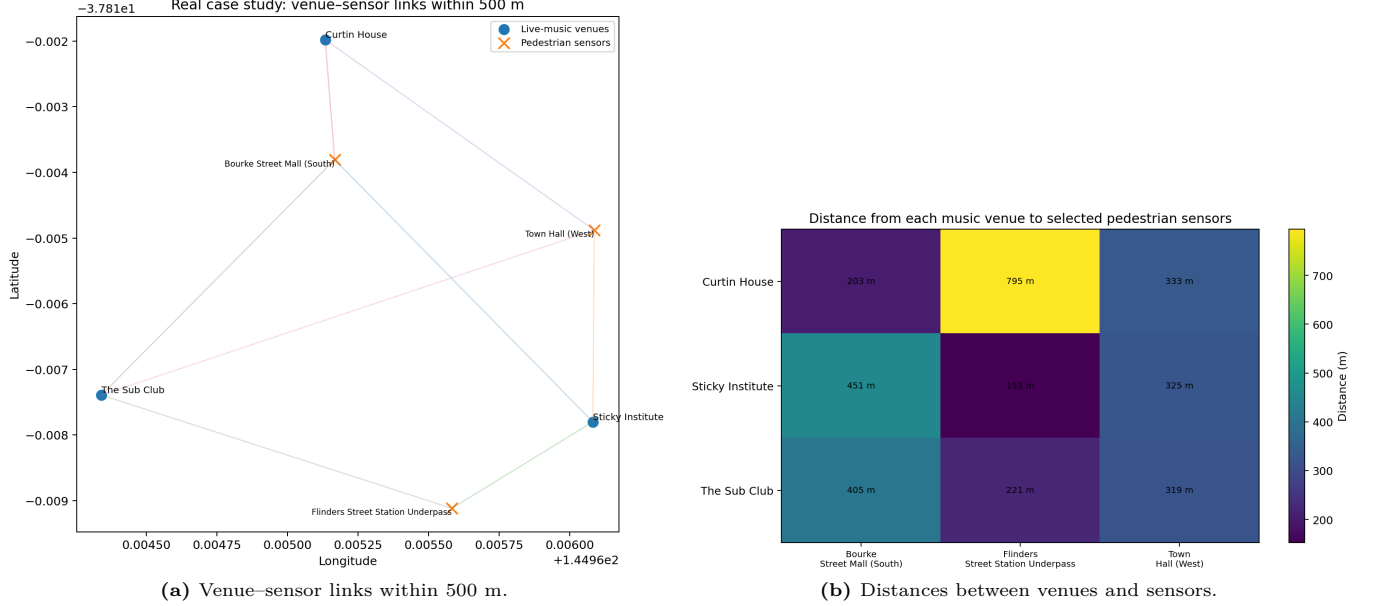


Figure 1: Spatial matching in the real Melbourne case study.

Figure 1a visualises the ambiguity of spatial matching. Each music venue can be associated with more than one nearby sensor, so a local footfall value depends on whether the analysis uses the nearest sensor, several nearby sensors, or a distance-weighted measure. Figure 1b makes the same structure quantitative. Sticky Institute is closest to Flinders Street Station Underpass, Curtin House to Bourke Street Mall (South), and The Sub Club again to Flinders Street Station Underpass, while several alternative matches remain within 500 m. This motivates testing more than one spatial scale.

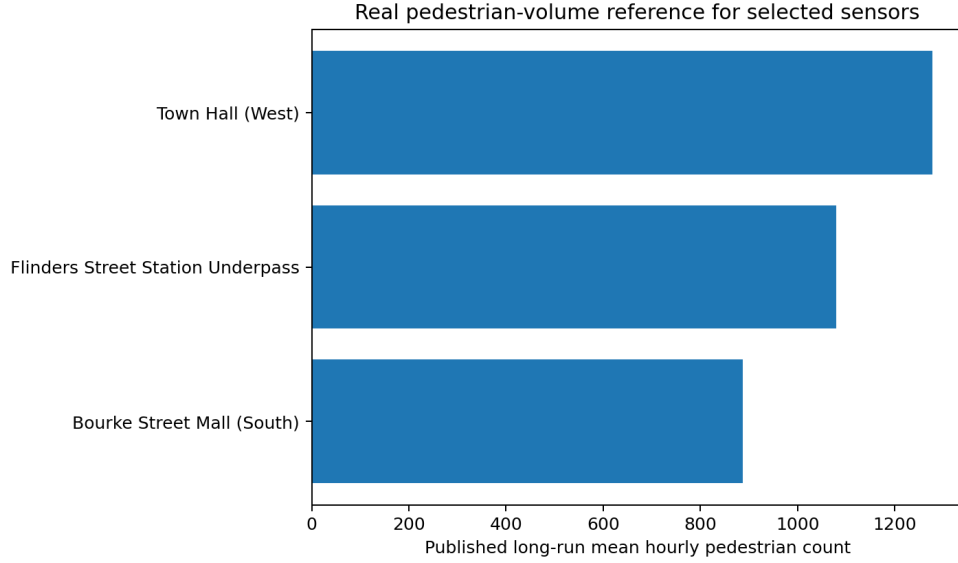


Figure 2: Reference mean hourly pedestrian volume for the selected sensors.

Figure 2 shows that nearby sensors can have substantially different baseline activity levels. Distance alone therefore does not guarantee that two sensors represent the same urban context.

4 Urban Activity and Consumption Context

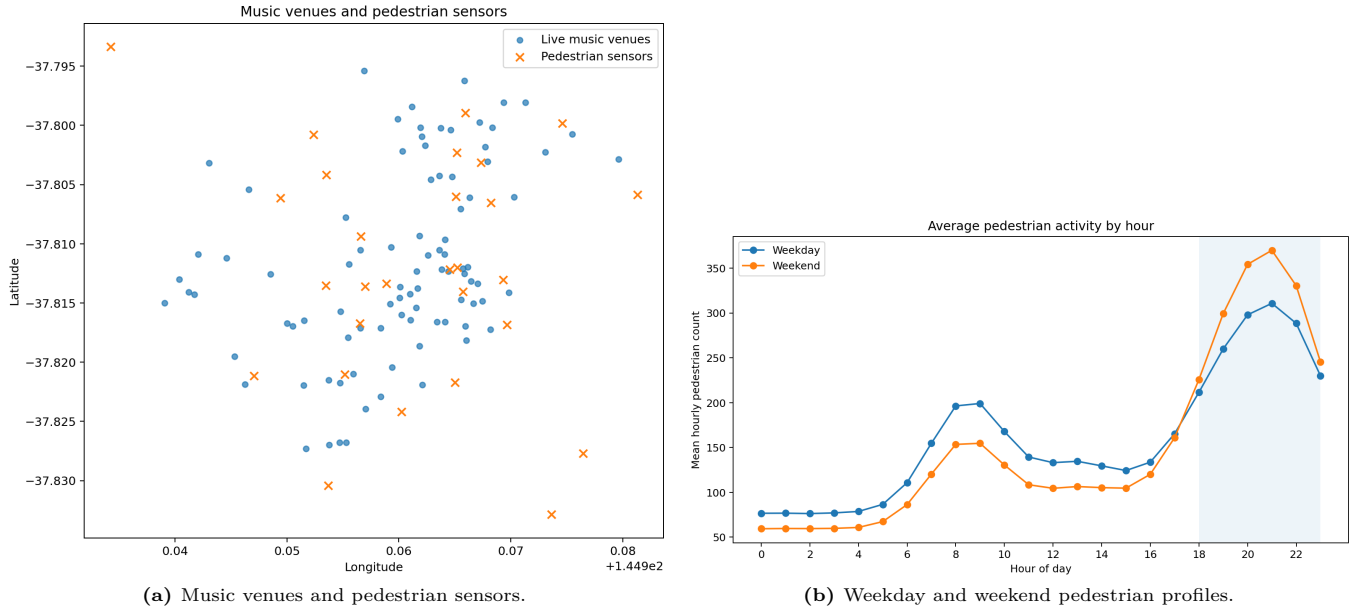


Figure 3: Spatial coverage and temporal patterns in the integrated city analysis.

Figure 3a shows how music locations are distributed relative to pedestrian monitoring points and provides the spatial basis for calculating local venue density. Figure 3b reveals a clear difference between weekday and weekend rhythms. Weekdays show a stronger commuting pattern, while weekend activity remains comparatively stronger during the evening, supporting the use of an evening window when examining music-related night-time activity.

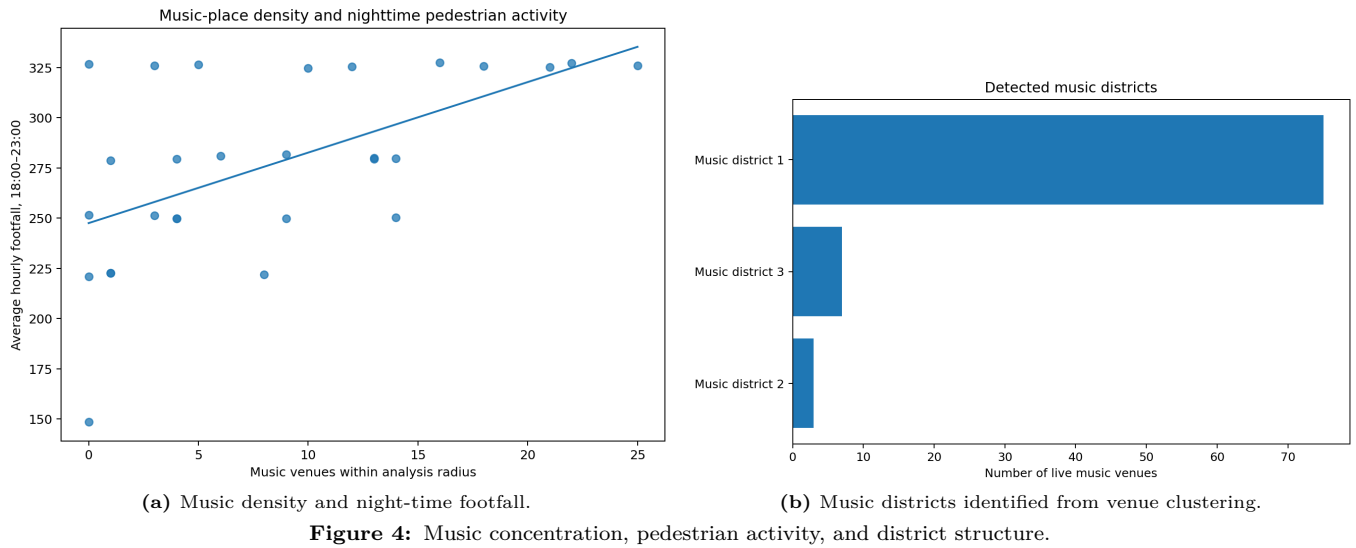


Figure 4a shows a positive relationship between local music density and night-time footfall in the integrated analysis. The pattern indicates that areas with more concentrated music activity also tend to experience greater pedestrian activity during the evening. Figure 4b converts individual venues into district-level clusters, making it possible to distinguish the main concentration of music activity from smaller secondary centres.

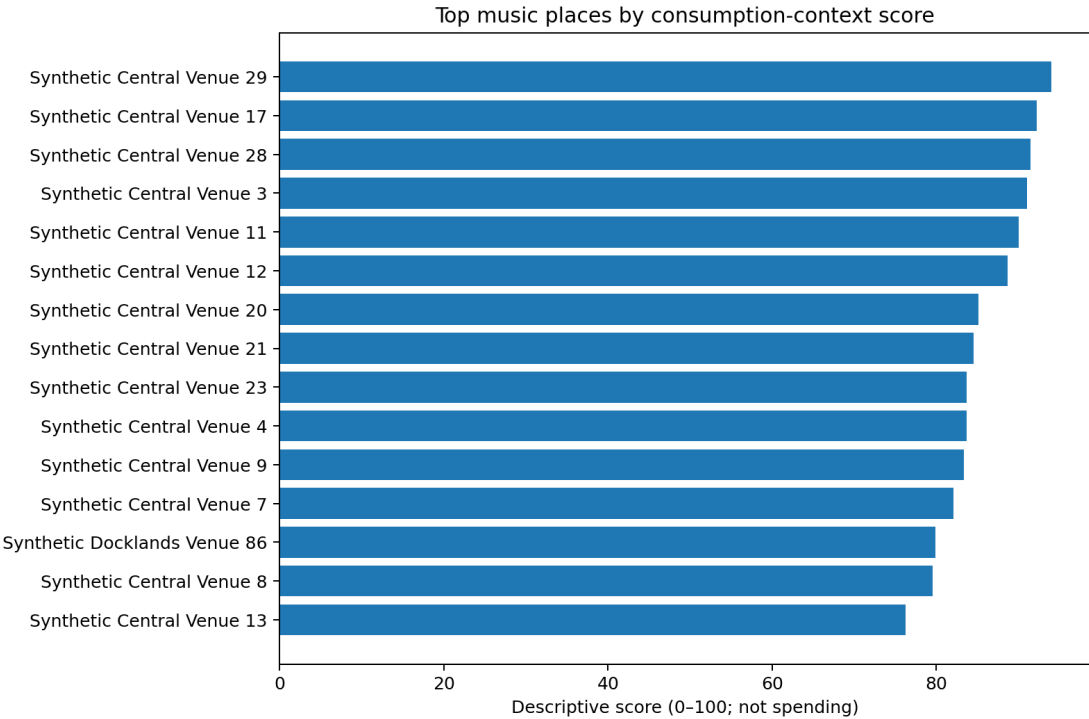


Figure 5: Highest descriptive consumption-context scores in the integrated analysis.

Figure 5 combines music density, night-time footfall, and nearby hospitality capacity into a descriptive consumption-context score. Higher values identify locations where cultural concentration, pedestrian activity, and consumption infrastructure coincide. The score therefore provides a compact way to compare the surrounding commercial context of different music locations without treating it as realised revenue.

5 Simulation Analysis

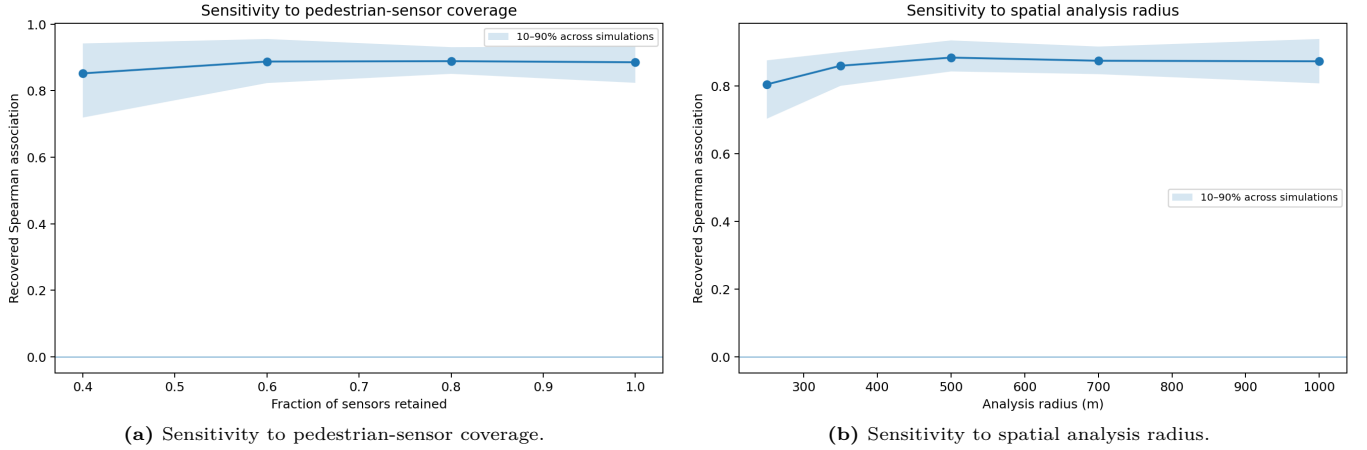


Figure 6: Effects of data coverage and spatial scale on the recovered association. Shading denotes the 10th–90th percentile range across simulation runs.

Figure 6a shows that lower sensor coverage mainly increases uncertainty. The mean association changes only moderately, from about 0.885 under full coverage to about 0.852 at 40% coverage, while the range across repeated runs becomes substantially wider. Figure 6b shows that a very small neighbourhood can miss relevant local structure. The recovered association rises from about 0.805 at 250 m to about 0.885 at 500 m and then changes relatively little at larger radii.

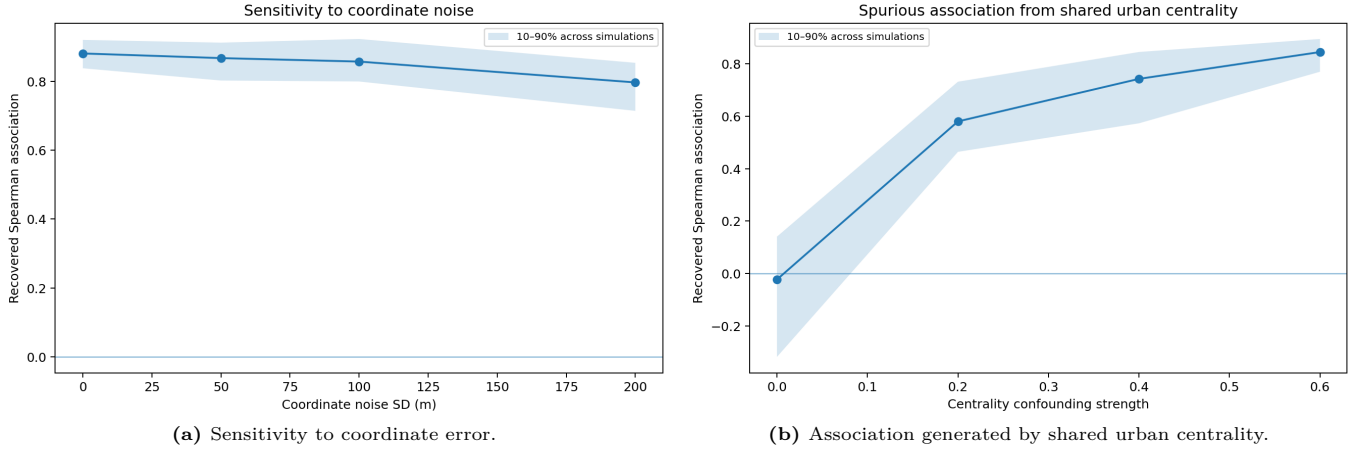


Figure 7: Measurement error and spatial confounding in the simulation analysis.

Figure 7a shows that location error weakens the recovered spatial relationship. The mean association falls from about 0.881 with no added error to about 0.797 under 200 m coordinate noise, indicating that local analysis becomes less reliable when venue or sensor positions are imprecise.

Figure 7b provides the most important interpretive result. When the direct music effect is fixed at zero, the recovered association rises from approximately -0.023 with no centrality confounding to 0.845 under strong shared centrality. Central locations can therefore produce high music density and high pedestrian activity at the same time, generating a strong observed relationship without a direct music effect.

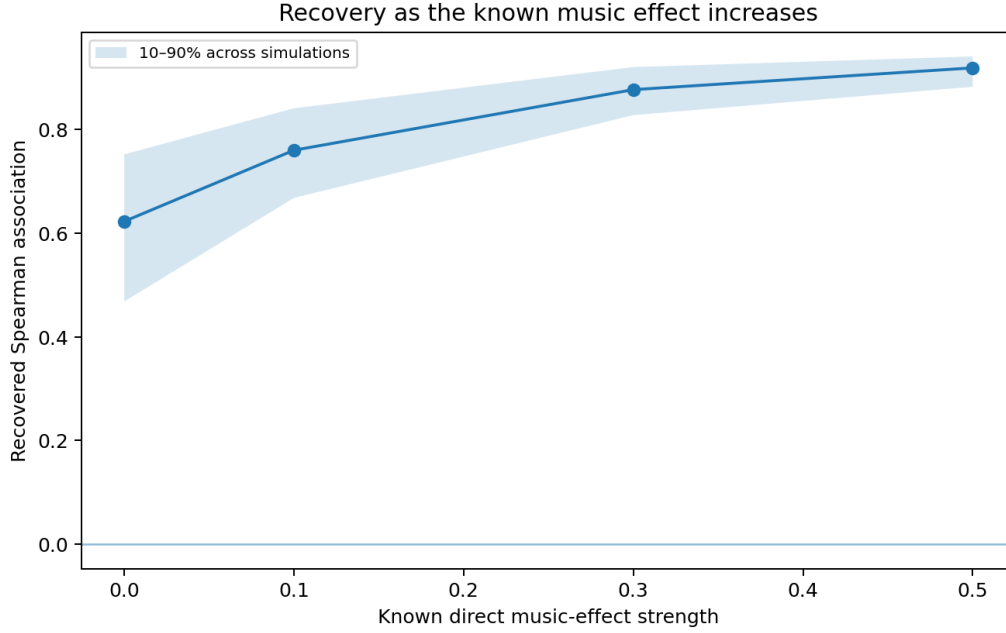


Figure 8: Recovered association as the programmed direct music effect increases under moderate centrality confounding.

Figure 8 shows that the method responds to a stronger direct signal. The mean recovered association increases from about 0.623 at zero direct effect to 0.919 at the largest tested effect. The positive starting value reflects the remaining centrality component, reinforcing the distinction between observed correlation and direct effect size.

Table 2: Key findings from the simulation analysis.

Diagnostic	Result	Interpretation
500 m spatial radius	$\bar{\rho} = 0.885$	The local relationship is recovered strongly around the main analysis scale.
200 m coordinate error	$\bar{\rho} : 0.881 \rightarrow 0.797$	Large location error weakens recovery of local spatial structure.
Zero direct music effect	$\bar{\rho} : -0.023 \rightarrow 0.845$	Shared centrality alone can generate a strong positive association.
Increasing direct effect	$\bar{\rho} : 0.623 \rightarrow 0.919$	The recovered association increases as the direct music effect becomes stronger.

6 Conclusion

MelodyMap examines how urban music-space concentration relates to night-time pedestrian activity and nearby hospitality capacity. The Melbourne case study shows that this relationship depends on spatial matching choices, since several pedestrian sensors can plausibly represent the same venue. The integrated analysis further shows that music-dense areas can coincide with stronger evening footfall and richer consumption environments, providing a practical way to identify and compare urban music districts. The simulation analysis clarifies how these patterns should be interpreted. Lower sensor coverage and coordinate error make the estimates less stable, while shared urban centrality can generate a strong music-footfall association even when the direct music effect is zero. The results therefore support the use of open urban data for mapping music-related night-time activity, while showing that spatial association alone is insufficient to establish a direct economic effect of music venues.